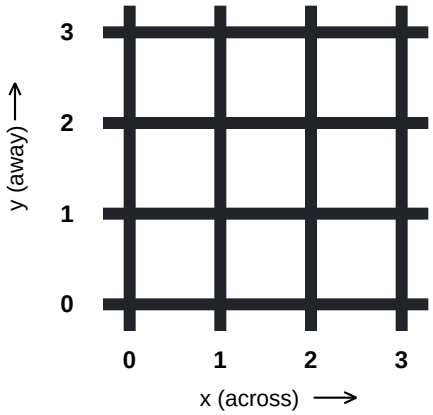
Lesson 4-3 · Trace it on paper

Name: _____ Date: _____

Trace by hand the six paths your tests will check. Across first, then up or down. Each move changes x or y by 1, never both. The start is never in the path.
When you run your code, this sheet is your answer key: every line on screen should match it.



Draw each path in pencil; erase it before the next.

1. Path 1, move by move: (0, 0) to (2, 3)

One row per move. Row 1 is done for you.

Step	Action	New position
1	x + 1	(1, 0)
2		
3		
4		
5		

2. All six paths

Write each expected path and its number of steps. Steps = |end x - start x| + |end y - start y|.

#	From / to	Expected path	Steps	Program matched?
`test_paper_paths()`				
1	(0, 0) to (2, 3)			Y / N
2	(0, 1) to (3, 3)			Y / N
3	(3, 3) to (1, 0)			Y / N
`test_edge_cases()`				
4	(2, 0) to (2, 3)			Y / N
5	(0, 2) to (3, 2)			Y / N
6	(1, 1) to (1, 1)			Y / N

Fill in Expected path and Steps now. Circle Y or N after you run your tests.

3. What is different about path 3 compared with paths 1 and 2?

(Look at which way x and y change.)